Emil Kasper

Vienna and Zurich kaspere@ethz.ch emil-kasper

Experience

Agile Robots SE, Working Student — Munich, Germany
- Designing and running machine-learning training environments. — Oct 2026 – present
- Independently researching and implementing human-in-the-loop reinforcement learning (HIL-RL) for dual-arm manipulation tasks. — 2 months

IVIA Lab, ETH Zurich, Research Intern — Zurich, Switzerland
Supervised by Prof. Menna El-Assady. — July 2026 – Aug 2026
- Selected as one of 20 students from over 4,000 applicants for the ETH Student Summer Research Fellowship. — 2 months
- Working on agentic AI and agent-agent communication.

Siemens, Working Student — Vienna, Austria
- Contributing to a test simulation framework for the European Train Control System (ETCS). — July 2025 – June 2026
- Implementing the autonomous train operation (ATO) mode in C# for test simulations. — 1 year

Institute for Visual Computing and Human-Centered Technology, TU Wien, Teaching Assistant — Vienna, Austria
Course: Interface and Interaction Design. — Mar 2025 – June 2026
- Guiding students through the lifecycle of user-centered product development. — 1 year 4 months
- Providing feedback and grading for 45 students.

Machine Learning Research Unit, TU Wien, Undergraduate Researcher — Vienna, Austria
Supervised by Prof. Clemens Heitzinger. — Aug 2025 – Apr 2026
- Proving better convergence and stability guarantees for Q-Learning and Speedy Q-Learning through batch sampling. — 9 months
- Experimenting with PPO for robotic hand manipulation.

Education

MSc — **ETH Zurich**, Computer Science — Zurich, Switzerland
- Major in Machine Intelligence. — Sept 2026 – June 2028

BSc (Honors) — **TU Wien (Vienna University of Technology)**, Computer Science — Vienna, Austria
- Four-year honors degree with a minor in machine learning. — Oct 2022 – May 2026
- Graduated with distinction.
- Received merit-based scholarships.

Research

Asking Your Advisor: Learning When to Consult Stronger Models in Agentic Systems
Submitted to ICLR 2027. We train a meta-policy that learns when to consult a stronger advisor model and when to continue with a cheaper one, reaching a better success-cost Pareto frontier than standard advisor-executor setups. Preprint on arXiv.

Student Engagements

European Student Robotics Association (ESRA), Executive Board Member, Partnerships — Europe
- Serving on the executive board of ESRA, the European Student Robotics Association. — Sept 2026 – present
- Driving partnerships to push the boundaries of European robotics.

TU Wien Robotics Club, Machine Learning Engineer — Vienna, Austria
Apr 2026 – Sept 2026

- Implementing VLAs for robotic applications with an SO101 arm.
- Building a chess robot that autonomously plays chess and moves the pieces.

TU Wien Space Team, Software Developer — Vienna, Austria
Mar 2025 – Oct 2025

- Building frontend software for the control system of Hedy, the liquid-fuel rocket of the Lamarr project, at the 2025 European Rocketry Challenge (EuRoC).
- Hedy reached 5,336 meters and broke the sound barrier at Mach 1.1, the highest flight in the Liquid 9 km category and the second highest of the entire competition (launch video).

Programs

TU Wien Informatics, Bachelor with Honors — Vienna, Austria
Oct 2024 – May 2026

- Selected for the Informatics excellence program, a performance-based honors track comparable to the Anglo-American honors model.
- Extends the studies by one year with research-oriented and master's-level courses (program website).

European Student Assembly 2026, EUC Voices, Delegate — Strasbourg, France
Jan 2026 – Apr 2026

- Representing the EULiST alliance at the European Student Assembly 2026.
- Writing policy recommendations on empowering inclusive digital literacy and building cyber-resilience at a European level.

Languages

German
Native

English
Fluent

Russian
Fluent